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chcp 65001
SET PGCLIENTENCODING=UTF8
SET PAGER=
psql -U postgres -d beer_import_db -f queries.sql
psql -U postgres -d beer_import_db -f indexes.sql
```

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SET
SET
---1---

QUERY PLAN
-----
Sort (cost=3302.60..3305.11 rows=1005 width=181) (actual time=152.553..152.591 rows=1005.00 loops=1)
  Sort Key: (COALESCE(s.sales_rub, '0'::numeric)) DESC
  Sort Method: quicksort Memory: 105kB
  Buffers: shared hit=2859
  -> Hash Left Join (cost=481.47..3252.48 rows=1005 width=181) (actual time=50.149..151.558 rows=1005.00 loops=1)
    Hash Cond: (b.id = st.beer_id)
    Buffers: shared hit=2856
    -> Hash Left Join (cost=263.14..2998.84 rows=1005 width=85) (actual time=35.846..135.761 rows=1005.00 loops=1)
      Hash Cond: (b.id = s.beer_id)
      Buffers: shared hit=2807
      -> Seq Scan on beer b (cost=0.00..2733.05 rows=1005 width=21) (actual time=0.105..94.459 rows=1005.00 loops=1)
        Buffers: shared hit=2723
      -> Hash (cost=259.80..259.80 rows=267 width=68) (actual time=35.719..35.726 rows=228.00 loops=1)
        Buckets: 1024 Batches: 1 Memory Usage: 20kB
        Buffers: shared hit=84
        -> Subquery Scan on s (cost=253.13..259.80 rows=267 width=68) (actual time=35.610..35.688 rows=228.00 loops=1)
          Buffers: shared hit=84
          -> HashAggregate (cost=253.13..257.13 rows=267 width=68) (actual time=35.609..35.672 rows=228.00 loops=1)
            Group Key: tdi.beer_id
            Batches: 1 Memory Usage: 304kB
            Buffers: shared hit=84
            -> Hash Join (cost=12.29..250.46 rows=267 width=15) (actual time=0.104..34.592 rows=260.00 loops=1)
              Hash Cond: (tdi.document_id = td.id)
              Buffers: shared hit=84
              -> Seq Scan on transfer_document_item tdi (cost=0.00..205.27 rows=12527 width=19) (actual time=0.024..21.642 rows=12527.00 loops=1)
                Buffers: shared hit=80
              -> Hash (cost=8.96..8.96 rows=267 width=4) (actual time=0.073..0.073 rows=260.00 loops=1)
                Buckets: 1024 Batches: 1 Memory Usage: 18kB
                Buffers: shared hit=4
                -> Index Only Scan using idx_td_transfer_date_id on transfer_document td (cost=0.29..8.96 rows=267 width=4) (actual time=0.023..0.048 rows=260.00 loops=1)
                  Index Cond: (transfer_date >= '2026-01-01'::date)
                  Heap Fetches: 0
                  Index Searches: 1
                  Buffers: shared hit=4
            -> Hash (cost=205.78..205.78 rows=1004 width=36) (actual time=14.288..14.289 rows=1004.00 loops=1)
              Buckets: 1024 Batches: 1 Memory Usage: 52kB
              Buffers: shared hit=49
              -> Subquery Scan on st (cost=183.19..205.78 rows=1004 width=36) (actual time=13.959..14.190 rows=1004.00 loops=1)
                Buffers: shared hit=49
                -> HashAggregate (cost=183.19..195.74 rows=1004 width=36) (actual time=13.958..14.127 rows=1004.00 loops=1)
                  Group Key: warehouse_stock.beer_id
                  Batches: 1 Memory Usage: 625kB
                  Buffers: shared hit=49
                  -> Seq Scan on warehouse_stock (cost=0.00..138.46 rows=8946 width=8) (actual time=0.033..6.757 rows=8946.00 loops=1)
                    Buffers: shared hit=49
Planning:
  Buffers: shared hit=306
Planning Time: 133.013 ms
Execution Time: 156.649 ms
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QUERY PLAN

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Sort (cost=798.73..799.40 rows=267 width=172) (actual time=5.439..5.449 rows=118.00 loops=1)
  Sort Key: (sum(sd.day_sales_rub)) DESC
  Sort Method: quicksort Memory: 35kB
  Buffers: shared hit=292
  CTE sales_detail
    -> HashAggregate (cost=664.70..668.71 rows=267 width=72) (actual time=4.020..4.112 rows=247.00 loops=1)
      Group Key: dc.venue_id, td.transfer_date
      Batches: 1 Memory Usage: 304kB
      Buffers: shared hit=256
      -> Hash Join (cost=481.43..661.36 rows=267 width=19) (actual time=2.876..3.795 rows=260.00 loops=1)
        Hash Cond: (dc.id = td.contract_id)
        Buffers: shared hit=256
        -> Seq Scan on delivery_contract dc (cost=0.00..146.01 rows=5001 width=8) (actual time=0.016..0.434 rows=5001.00 loops=1)
          Buffers: shared hit=96
        -> Hash (cost=478.09..478.09 rows=267 width=19) (actual time=2.850..2.852 rows=260.00 loops=1)
          Buckets: 1024 Batches: 1 Memory Usage: 22kB
          Buffers: shared hit=160
          -> Hash Join (cost=239.93..478.09 rows=267 width=19) (actual time=1.216..2.805 rows=260.00 loops=1)
            Hash Cond: (tdi.document_id = td.id)
            Buffers: shared hit=160
            -> Seq Scan on transfer_document_item tdi (cost=0.00..205.27 rows=12527 width=15) (actual time=0.009..0.641 rows=12527.00 loops=1)
              Buffers: shared hit=80
            -> Hash (cost=236.59..236.59 rows=267 width=12) (actual time=1.198..1.198 rows=260.00 loops=1)
              Buckets: 1024 Batches: 1 Memory Usage: 20kB
              Buffers: shared hit=80
              -> Seq Scan on transfer_document td (cost=0.00..236.59 rows=267 width=12) (actual time=0.015..1.156 rows=260.00 loops=1)
                Filter: (transfer_date >= '2026-01-01':date)
                Rows Removed by Filter: 12267
                Buffers: shared hit=80
          -> HashAggregate (cost=114.59..119.27 rows=267 width=172) (actual time=5.278..5.345 rows=118.00 loops=1)
            Group Key: v.id, gf.has_long_gap
            Batches: 1 Memory Usage: 176kB
            Buffers: shared hit=289
            -> Hash Left Join (cost=105.18..111.92 rows=267 width=108) (actual time=4.993..5.137 rows=247.00 loops=1)
              Hash Cond: (v.id = gf.venue_id)
              Buffers: shared hit=289
              -> Hash Join (cost=72.07..78.11 rows=267 width=107) (actual time=4.569..4.661 rows=247.00 loops=1)
                Hash Cond: (sd.venue_id = v.id)
                Buffers: shared hit=283
                -> CTE Scan on sales_detail sd (cost=0.00..5.34 rows=267 width=68) (actual time=4.023..4.044 rows=247.00 loops=1)
                  Storage: Memory Maximum Storage: 28kB
                  Buffers: shared hit=256
                -> Hash (cost=47.03..47.03 rows=2003 width=43) (actual time=0.538..0.538 rows=2003.00 loops=1)
                  Buckets: 2048 Batches: 1 Memory Usage: 167kB
                  Buffers: shared hit=27
                  -> Seq Scan on venue v (cost=0.00..47.03 rows=2003 width=43) (actual time=0.015..0.248 rows=2003.00 loops=1)
                    Buffers: shared hit=27
              -> Hash (cost=30.61..30.61 rows=200 width=5) (actual time=0.416..0.418 rows=118.00 loops=1)
                Buckets: 1024 Batches: 1 Memory Usage: 13kB
                Buffers: shared hit=6
                -> Subquery Scan on gf (cost=16.10..30.61 rows=200 width=5) (actual time=0.241..0.400 rows=118.00 loops=1)
                  Buffers: shared hit=6
                  -> GroupAggregate (cost=16.10..28.61 rows=200 width=5) (actual time=0.241..0.390 rows=118.00 loops=1)
                    Group Key: sales_detail.venue_id
                    Buffers: shared hit=6
                    -> WindowAgg (cost=16.10..22.11 rows=267 width=12) (actual time=0.235..0.348 rows=247.00 loops=1)
                      Window: w1 AS (PARTITION BY sales_detail.venue_id ORDER BY sales_detail.transfer_date)
                      Storage: Memory Maximum Storage: 17kB
                      Buffers: shared hit=6
                    -> Sort (cost=16.10..16.77 rows=267 width=8) (actual time=0.225..0.235 rows=247.00 loops=1)
                      Sort Key: sales_detail.venue_id, sales_detail.transfer_date
                      Sort Method: quicksort Memory: 30kB
                      Buffers: shared hit=6
                      -> CTE Scan on sales_detail (cost=0.00..5.34 rows=267 width=8) (actual time=0.001..0.151 rows=247.00 loops=1)
                        Storage: Memory Maximum Storage: 28kB
    Planning:
      Buffers: shared hit=262
    Planning Time: 1.776 ms
    Execution Time: 5.786 ms
(69 rows)
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QUERY PLAN
-----
Sort (cost=5230.75..5233.27 rows=1005 width=181) (actual time=7.600..7.643 rows=1005.00 loops=1)
  Sort Key: (COALESCE(s.sales_rub, '0')::numeric) DESC
  Sort Method: quicksort Memory: 105kB
  Buffers: shared hit=1515 read=52
  -> Hash Left Join (cost=579.68..5180.64 rows=1005 width=181) (actual time=5.774..7.201 rows=1005.00 loops=1)
    Hash Cond: (b.id = s.beer_id)
    Buffers: shared hit=1515 read=52
    -> Hash Left Join (cost=218.61..4784.25 rows=1005 width=53) (actual time=3.837..4.808 rows=1005.00 loops=1)
      Hash Cond: (b.id = st.beer_id)
      Buffers: shared hit=1042
      -> Index Scan using beer_pkey on beer b (cost=0.28..4563.27 rows=1005 width=21) (actual time=0.018..0.724 rows=1005.00 loops=1)
        Index Searches: 1
        Buffers: shared hit=993
      -> Hash (cost=205.78..205.78 rows=1004 width=36) (actual time=3.807..3.808 rows=1004.00 loops=1)
        Buckets: 1024 Batches: 1 Memory Usage: 52kB
        Buffers: shared hit=49
        -> Subquery Scan on st (cost=183.19..205.78 rows=1004 width=36) (actual time=3.373..3.685 rows=1004.00 loops=1)
          Buffers: shared hit=49
          -> HashAggregate (cost=183.19..195.74 rows=1004 width=36) (actual time=3.371..3.597 rows=1004.00 loops=1)
            Group Key: warehouse_stock.beer_id
            Batches: 1 Memory Usage: 625kB
            Buffers: shared hit=49
            -> Seq Scan on warehouse_stock (cost=0.00..138.46 rows=8946 width=8) (actual time=0.011..0.578 rows=8946.00 loops=1)
              Disabled: true
              Buffers: shared hit=49
          -> Hash (cost=357.74..357.74 rows=267 width=68) (actual time=1.911..1.912 rows=228.00 loops=1)
            Buckets: 1024 Batches: 1 Memory Usage: 20kB
            Buffers: shared hit=473 read=52
            -> Subquery Scan on s (cost=351.06..357.74 rows=267 width=68) (actual time=1.782..1.877 rows=228.00 loops=1)
              Buffers: shared hit=73 read=52
              -> HashAggregate (cost=351.06..355.07 rows=267 width=68) (actual time=1.781..1.857 rows=228.00 loops=1)
                Group Key: tdi.beer_id
                Batches: 1 Memory Usage: 304kB
                Buffers: shared hit=473 read=52
                -> Nested Loop (cost=0.57..348.39 rows=267 width=15) (actual time=0.192..1.591 rows=260.00 loops=1)
                  Buffers: shared hit=473 read=52
                  -> Index Only Scan using idx_td_transfer_date_id on transfer_document td (cost=0.29..8.96 rows=267 width=4) (actual time=0.169..0.205 rows=260.00 loops=1)
                    Index Cond: (transfer_date >= '2026-01-01'::date)
                    Heap Fetches: 0
                    Index Searches: 1
                    Buffers: shared hit=1 read=3
                  -> Index Only Scan using idx_document_beer on transfer_document_item tdi (cost=0.29..1.26 rows=1 width=19) (actual time=0.005..0.005 rows=1.00 loops=260)
                    Index Cond: (document_id = td.id)
                    Heap Fetches: 0
                    Index Searches: 260
                    Buffers: shared hit=472 read=49
Planning:
  Buffers: shared hit=158 read=6
Planning Time: 2.521 ms
Execution Time: 7.921 ms
(50 rows)

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--5--
QUERY PLAN
-----
Sort (cost=869.67..870.34 rows=267 width=172) (actual time=3.093..3.104 rows=118.00 loops=1)
  Sort Key: (sum(sd.day_sales_rub)) DESC
  Sort Method: quicksort Memory: 35kB
  Buffers: shared hit=734
  CTE sales_detail
  -> HashAggregate (cost=698.03..702.04 rows=267 width=72) (actual time=1.824..1.919 rows=247.00 loops=1)
    Group Key: dc.venue_id, td.transfer_date
    Batches: 1 Memory Usage: 304kB
    Buffers: shared hit=696
    -> Nested Loop (cost=101.02..694.70 rows=267 width=19) (actual time=0.200..1.623 rows=260.00 loops=1)
      Buffers: shared hit=696
      -> Merge Join (cost=100.74..355.26 rows=267 width=12) (actual time=0.190..1.217 rows=260.00 loops=1)
        Merge Cond: (td.contract_id = dc.id)
        Buffers: shared hit=175
        -> Sort (cost=100.45..101.12 rows=267 width=12) (actual time=0.178..0.195 rows=260.00 loops=1)
          Sort Key: td.contract_id
          Sort Method: quicksort Memory: 33kB
          Buffers: shared hit=65
          -> Bitmap Heap Scan on transfer_document td (cost=6.35..89.69 rows=267 width=12) (actual time=0.042..0.151 rows=260.00 loops=1)
            Recheck Cond: (transfer_date >= '2026-01-01'::date)
            Heap Blocks: exact=62
            Buffers: shared hit=65
            -> Bitmap Index Scan on idx_td_transfer_date_id (cost=0.00..6.29 rows=267 width=0) (actual time=0.021..0.021 rows=260.00 loops=1)
              Index Cond: (transfer_date >= '2026-01-01'::date)
              Index Searches: 1
              Buffers: shared hit=3
            -> Index Scan using delivery_contract_pkey on delivery_contract dc (cost=0.28..238.30 rows=5001 width=8) (actual time=0.009..0.675 rows=4963.00 loops=1)
              Index Searches: 1
              Buffers: shared hit=110
          -> Index Only Scan using idx_document_beer on transfer_document_item tdi (cost=0.29..1.26 rows=1 width=15) (actual time=0.001..0.001 rows=1.00 loops=260)
            Index Cond: (document_id = td.id)
            Heap Fetches: 0
            Index Searches: 260
            Buffers: shared hit=521
      -> HashAggregate (cost=152.20..156.87 rows=267 width=172) (actual time=2.995..3.052 rows=118.00 loops=1)
        Group Key: v.id, ((sum((sales_detail.transfer_date - lag(sales_detail.transfer_date) OVER w1))) > 30)
        Batches: 1 Memory Usage: 176kB
        Buffers: shared hit=734
        -> Merge Left Join (cost=32.48..149.53 rows=267 width=100) (actual time=2.182..2.874 rows=247.00 loops=1)
          Merge Cond: (v.id = sales_detail.venue_id)
          Buffers: shared hit=734
          -> Merge Join (cost=16.38..117.48 rows=267 width=107) (actual time=2.097..2.578 rows=247.00 loops=1)
            Merge Cond: (v.id = sd.venue_id)
            Buffers: shared hit=734
            -> Index Scan using venue_pkey on venue v (cost=0.28..92.37 rows=2003 width=43) (actual time=0.020..0.314 rows=1999.00 loops=1)
              Index Searches: 1
              Buffers: shared hit=38
            -> Sort (cost=16.10..16.77 rows=267 width=68) (actual time=2.074..2.086 rows=247.00 loops=1)
              Sort Key: sd.venue_id
              Sort Method: quicksort Memory: 34kB
              Buffers: shared hit=696
              -> CTE Scan on sales_detail sd (cost=0.00..5.34 rows=267 width=68) (actual time=1.827..1.985 rows=247.00 loops=1)
                Storage: Memory Maximum Storage: 28kB
                Buffers: shared hit=696
              -> GroupAggregate (cost=16.10..28.61 rows=200 width=5) (actual time=0.084..0.250 rows=118.00 loops=1)
                Group Key: sales_detail.venue_id
                -> WindowAgg (cost=16.10..22.11 rows=267 width=12) (actual time=0.079..0.198 rows=247.00 loops=1)
                  Window: w1 AS (PARTITION BY sales_detail.venue_id ORDER BY sales_detail.transfer_date)
                  Storage: Memory Maximum Storage: 17kB
                  -> Sort (cost=16.10..16.77 rows=267 width=8) (actual time=0.068..0.081 rows=247.00 loops=1)
                    Sort Method: quicksort Memory: 30kB
                    -> CTE Scan on sales_detail (cost=0.00..5.34 rows=267 width=8) (actual time=0.001..0.024 rows=247.00 loops=1)
                      Storage: Memory Maximum Storage: 28kB
Planning:
  Buffers: shared hit=94
Planning Time: 2.193 ms
Execution Time: 3.338 ms
(68 rows)

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|                | Запрос 1 без               | Запрос 1 с                            | Запрос 5 без                         | Запрос 5 с                                              |
|----------------|----------------------------|---------------------------------------|--------------------------------------|---------------------------------------------------------|
| Planning time  | 133.013 ms                 | 2.521 ms                              | 2.413 ms                             | 1.195 ms                                                |
| Execution time | 156.649 ms                 | 7.921 ms                              | 6.808 ms                             | 4.247 ms                                                |
| Scan types     | Bitmap Heap Scan, Seq Scan | Index Only Scan, Index Scan, Seq Scan | Bitmap Heap Scan, CTE Scan, Seq Scan | Bitmap Heap Scan, CTE Scan, Index Only Scan, Index Scan |
| JOINS          | Hash Join, Hash Left Join  | Hash Left Join, Nested Loop           | Hash Join, Hash Left Join            | Merge Join, Merge Left Join, Nested Loop                |